

Paul Lee

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EDUCATION

The University of Texas at Austin, Austin, TX

Expected May 2028

Bachelor of Science, Mechanical Engineering | GPA 3.82

Certificate in Programming and Computation

Relevant Coursework: Thermodynamics, Dynamics, Statics, Engineering Computing Methods, Software Design, Elements of Database, and Electricity and Magnetism

WORK EXPERIENCE

Celestica - Dallas, TX

May 2026 - August 2026

Mechanical Engineering Intern

- Redesigned a sheet metal fan tray with a custom sliding mechanism, saving ~3.5 metric tons of steel (~\$10K in raw material cost) across a 60K-unit production run
- Designed sheet metal chassis and PCBA system component placements, reducing assembly time by up to 50% across AI server assemblies
- Revised 150+ part and assembly drawings, applying GD&T and ballooning inspection-critical dimensions
- Supported validation of blades through thermal, leak, seismic, tilt, and roll testing to verify product performance
- Performed Worst case and RSS tolerance stackup analysis and global interference checks across large TLA assemblies

MINIMAX Lab - Austin, TX

Sept 2025 - Feb 2026

Research Assistant

- Modeled a 125 μm silicon AFM cantilever in COMSOL, extracting stress distribution, spring constant, and the first three flexural modes for probe calibration
- Reviewed literature on active AFM cantilevers and piezoelectric actuation to inform model parameters

DUCK Lab - Austin, TX

Oct 2024 - May 2025

Research Assistant

- Increased annual energy yield by 8.2% and profit by 40.7% at fixed module count and installed cost through layout optimization
- Built and benchmarked 50+ solar farm layouts in Aladdin, cutting design generation time by 11x times
- Ran controlled tests isolating tree count, building placement, and polygon geometry, producing design guidelines the lab now uses

OnRamps - Austin, TX

Sept 2025 - Dec 2025

Course Grader

- Consistently graded over 500 student submissions per week with full on-time completion across multiple sections

PROJECTS

Hero Robot

- Redesigned the robot chassis in CAD, correcting mounting hole misalignments across over half of the assembly
- Designed and prototyped rounded skirt geometries that reduced collision forces by ~40%
- Redimensioned chassis components such as the belly pan and box tube to ensure consistent modular fit

Crates and 1v1 Arena

- Designed crates and 1v1 arena models, integrating components sourced from McMaster-Carr for precise assembly
- Fabricated wooden barriers for competition using miter saws, planers, and drills
- Built a 50 ft x 50 ft PVC testing arena from CAD designs, reducing robot testing setup time by ~30%

Robotathon

- Assembled a functional robot that can identify colors, follow lines of any shapes and navigate through walls
- Fabricated custom chassis components by 3D printing the battery mount and the IR sensor mount for the chassis

SKILLS

Software: Solidworks, Creo, ANSYS, COMSOL, Python, PLM, EZ-tol, Aladdin, Excel

Technical: CAD, FEA, 2D Drawing, 3D Printing, DFMA, Tolerance Analysis, CFD, Woodworking, GD&T, Soldering,

Languages: Native in English and Korean

Certificates: CSWA, CornellX(FEA/CFD), CAM and Design Manufacturing(Fusion 360)

HONORS AND AWARDS

- University Honors - Fall 2024, Spring 2025
- 1st place in Robotathon 2024
- BC Achievement Scholarship Award